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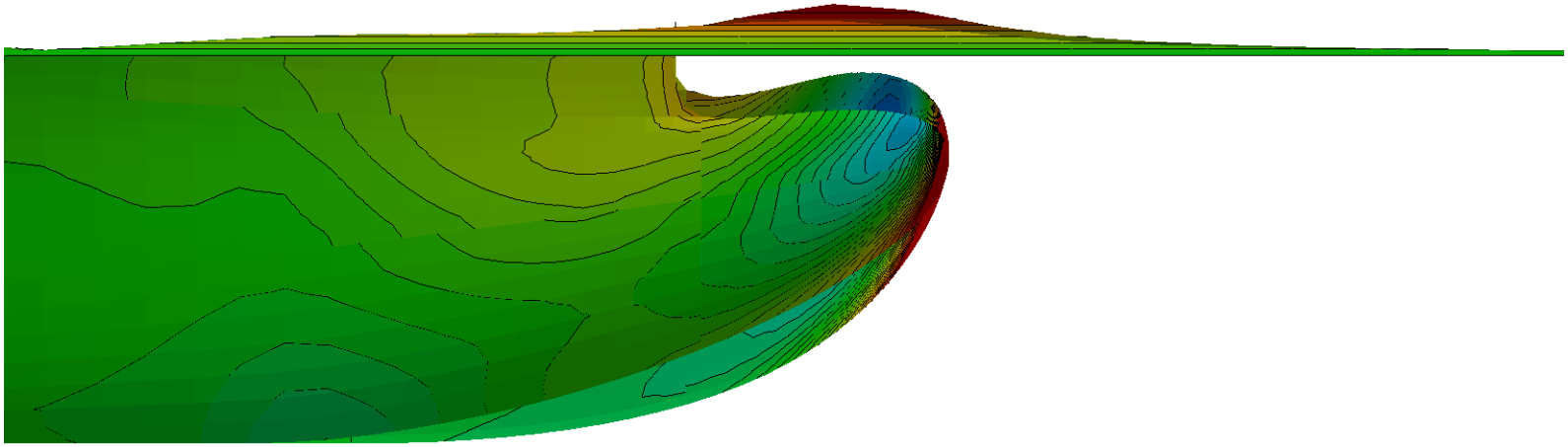

MARITIME

ECO Retrofit Potential for a 4,300 TEU container vessel

on behalf of Ahrenkiel Steamship GmbH & Co. KG

Dipl.-Ing. Juryk Henrichs

2014-12-02

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Keywords:

ECO-Retrofit Potential CV4300TEU

Abstract

For a 4,300 TEU container vessel two retrofit measures were assessed:

- A bulb refit study was carried out in order to identify the performance potential of a refitted bulbous bow.
  - The process combines a parametric geometry model of the bulbous bow region and advanced nonlinear hydrodynamic simulations based on potential theory to assess wave-pattern resistance and power consumption of the ship.
  - The sensitivity of the performance with respect to shape variations was analyzed and a representative candidate for improved performance on the operational profile identified.
- In addition the potentials from retrofitting a newly designed propeller have been analyzed for two scenarios – with and without de-rating measures on the main engine applied.

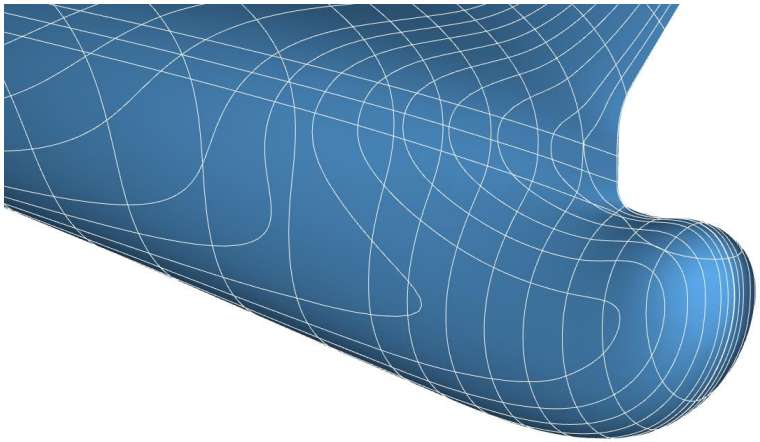
Content

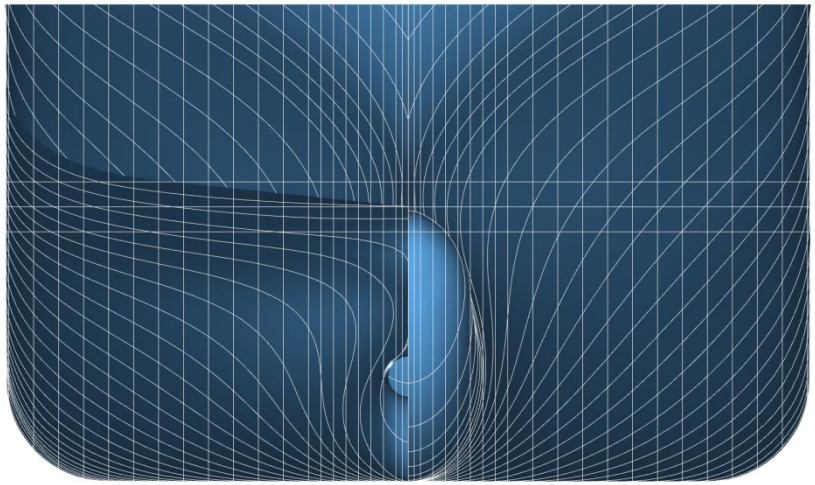
- Motivation
- General data and geometry of the vessel
  - Main particulars, impressions of baseline design
- *ECO Retrofit Potential Bow* – a bulb retrofit study
  - Bulbous bow region with cutout
  - Objectives
  - Operational profile
  - Parametric hull modeling and variation
  - Results
  - Flow field comparison of BL00.0 and exemplary design candidate
  - Impressions of an exemplary design candidate
  - Economical benefit estimation
- *ECO Retrofit Potential Propulsion*
  - Retrofit of a new propeller
  - Propulsion coefficients, efficiencies and drawings for existing and future propellers
  - Comparison of propeller main particulars
  - Conclusion
- Project summary

Motivation

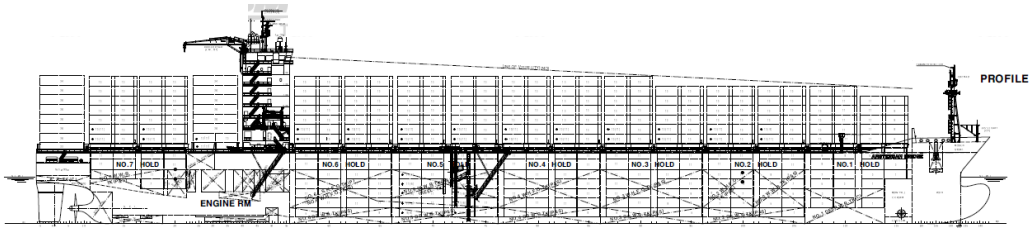
- In times of increasing fuel costs a strong focus of the economists of commercial shipping is the fuel consumption. This leads to a reduced optimum sailing speed as the impact of the fuel costs scales approximately with the 3rd power of the speed.
- Therefore the concept of slow steaming has been widely employed. However, many of today's ships have been designed with a strong focus on performance at design conditions and no emphasis was put on the off-design points which determine the performance while sailing in slow steaming conditions.
- A prominent feature determining the hydrodynamic performance of a vessel is the bulbous bow, and therefore, depending on the subject vessel, substantial potential may be available by a local modification.
- DNV GL has developed a dedicated procedure to identify available potentials by means of parametric modeling and fluid dynamic analysis. This allows to assess the potential of different variations in an efficient manner and may also be used to identify the most promising design.

Impressions of baseline hull geometry (BL00.0)





Baseline particulars:	
LOA	230.32 m
LPP	247.00 m
BOA	32.2 m
Design draught	11.0 m
Scantling draught	12.6 m
Main engine	MAN B&W 8K90MC-C6
$V_{service}$ @ T_{Des}	24.3 kts
Builder	DSME CO. LTD.
Year	≈2010



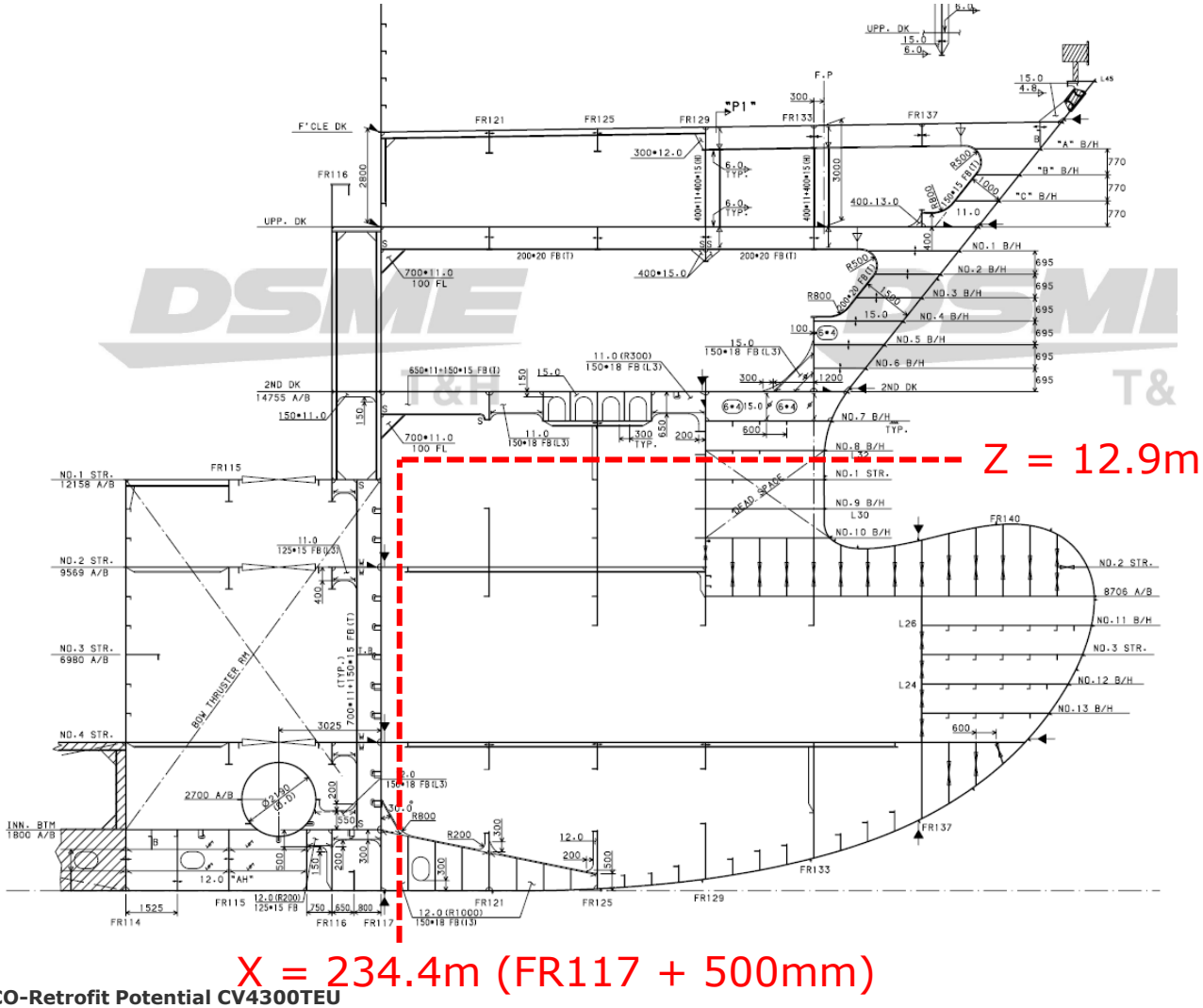
Courtesy Ahrenkiel Steamship GmbH & Co. KG

ECO-Retrofit Potential CV4300TEU

ECO Retrofit Potential - Bow

a bulb retrofit study

Bulbous bow region and cut out



- The horizontal cut out was chosen to be at z=12.9m
- The longitudinal position was chosen to be 0.5m ahead of the collision bulkhead at x = 234.4 m (FR117+0.5m)

Considered objectives

RTRWP

- Total resistance based on wave pattern resistance according to wave pattern analysis (to be minimized)

RT

- Total resistance based on wave making resistance (to be minimized)

Max crest to trough wave height

- Wave System along the ship hull (to be minimized)

Hydrodynamic fairness

- Measure for hydrodynamic fairness along the hull (to be minimized)

ITTC trial prediction of total resistance R_T

- $R_T = R_{W(P)} + (1+k)*R_F + R_A + R_{AA}$

- **Wave resistance**, dynamic floating position and assessment of the viscous resistance components by the state-of-the-art nonlinear Rankine panel code FS-Flow, which incorporates the wave pattern resistance from wave pattern analysis as additional information → providing R_{WP}

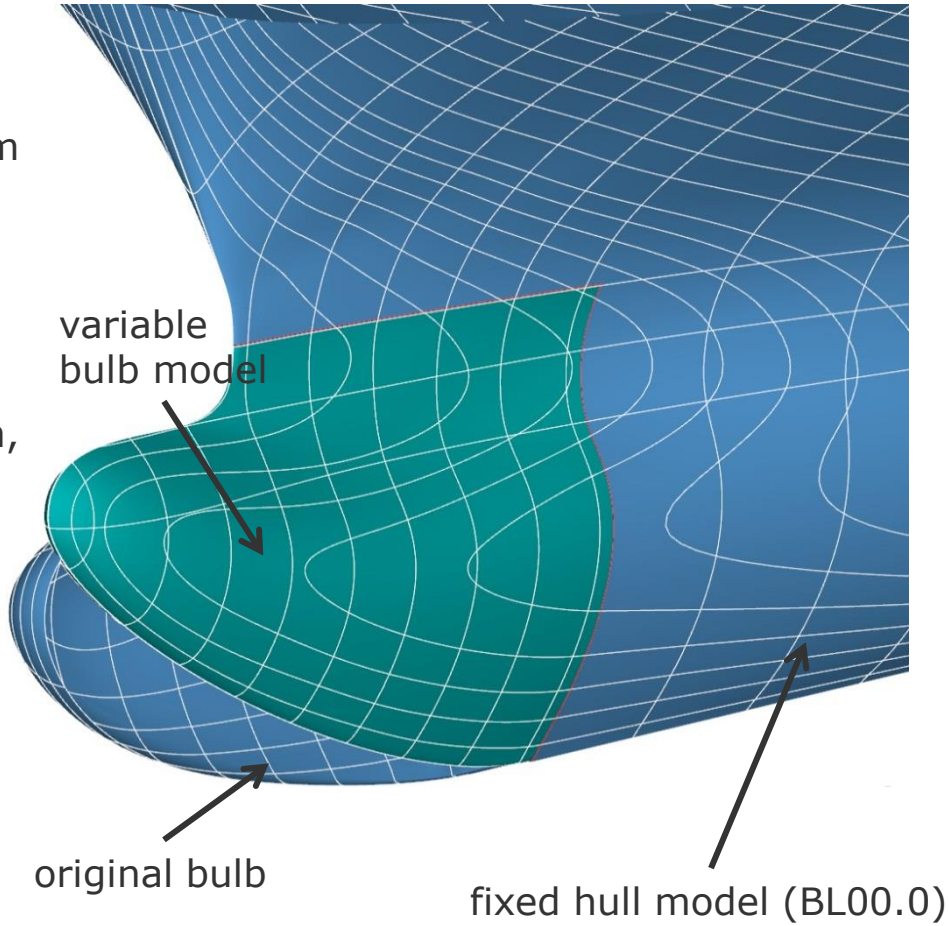
Operational profile

- Cases of operation are seen as an operational matrix.
  - Weightings are applied based on customers requirement.
  - Usually this is related to the estimated time and fuel oil consumption the vessel is sailing in each condition.

	A (20 kts)	B (18 kts)	C (16kts)	D (14 kts)
A (12.0 m)		AB (10 %)		AD (10 %)
B (11.0 m)	BA (5 %)	BB (20 %)	BC (40 %)	
C (10.0 m)				CD (15 %)

Parametric hull model

- For the bulb shape variation a fully parametric hull model was established.
- 19 free form parameters provide maximum shape flexibility.
- Integrated fairing algorithms ensure smooth surfaces for every single design variant.
- The main parameters are the bulb's length, tip height, height at Lpp, longitudinal und vertical volume distribution, entrance angles and some local fine tuning parameters.

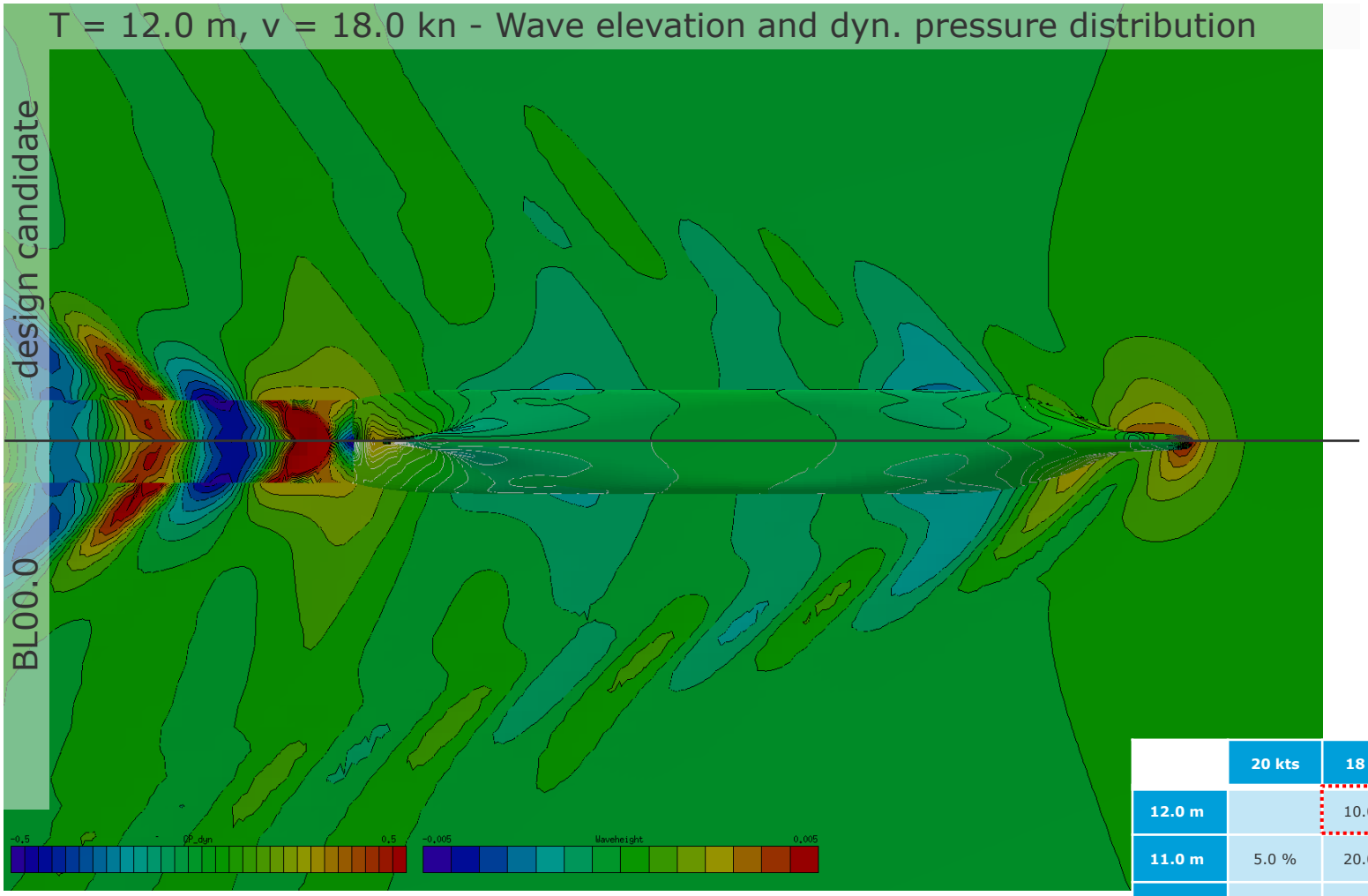


Results

Savings in Total Resistance R_T	A (20 kn)	B (18kn)	C (16 kn)	D (14 kn)
A (12.0 m)		1.0 %		6.5 %
B (11.0 m)	1.0 %	3.5 %	7.5 %	
C (10.0 m)				15.0 %

- Positive savings (green) denote an improved total resistance RT relative to the original geometry.
- This potential has been identified by means of an initial assessment based on potential theory and may be subject to small changes in a final evaluation with e.g. RANSE codes.
- An optimized bulbous bow will influence the propulsion efficiencies η_H and η_R (relative rotative propulsion efficiency) to some extent. However these effects were not considered in the present study and may be subjected to distinct model tests.
- The following slides show the flow details of the baseline design (BL00.0) and the improved flow field of an exemplary design candidate.
- The weighted saving across the operational profile sums up to **~6.8 %**.

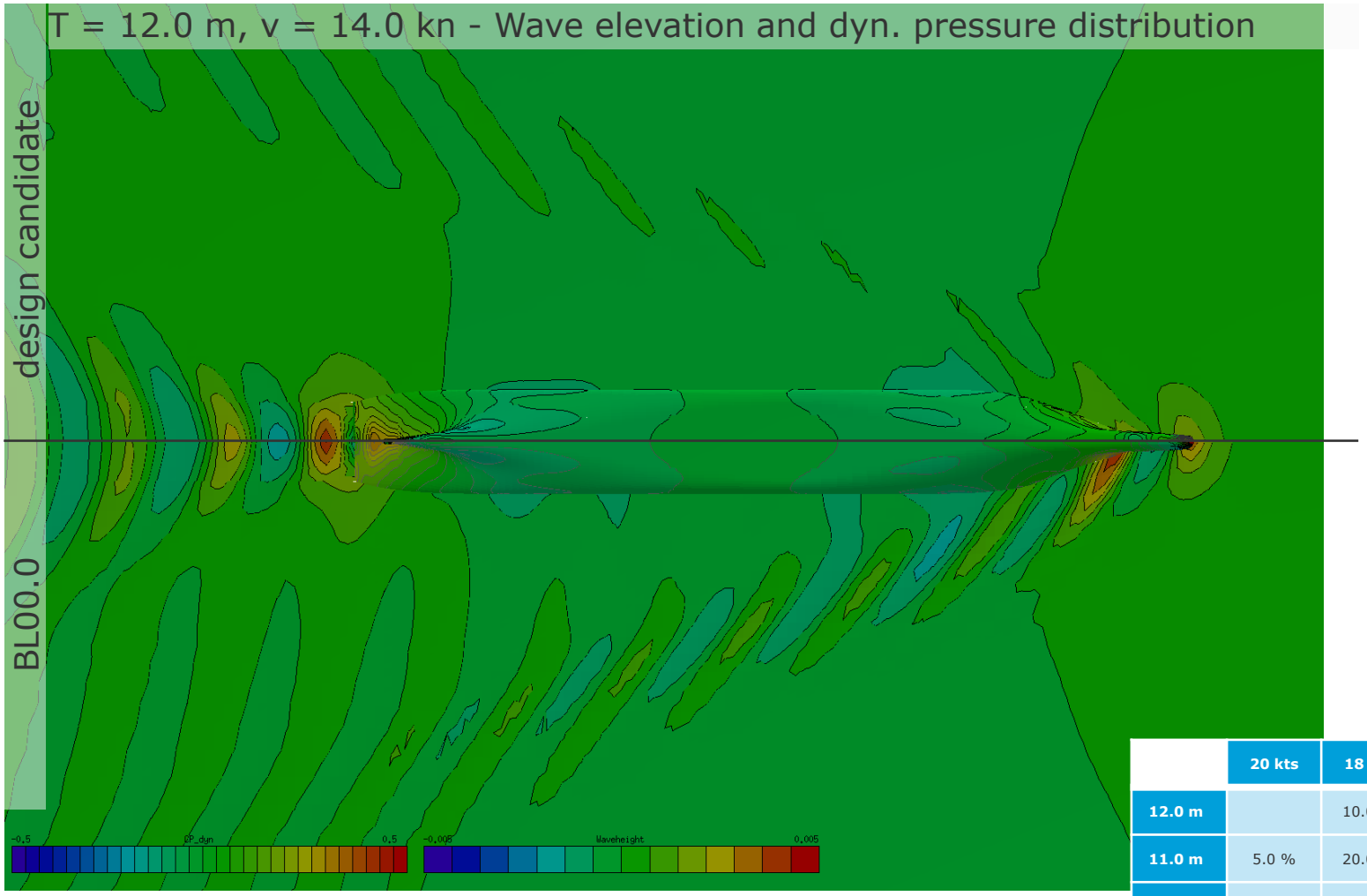
Flow field comparison of BL00.0 and exemplary design candidate



	20 kts	18 kts	16 kts	14 kts
12.0 m		10.0 %		10.0 %
11.0 m	5.0 %	20.0 %	40.0 %	
10.0 m				15.0 %

ECO-Retrofit Potential CV4300TEU

Flow field comparison of BL00.0 and exemplary design candidate



	20 kts	18 kts	16 kts	14 kts
12.0 m		10.0 %		10.0 %
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10.0 m				15.0 %

ECO-Retrofit Potential CV4300TEU

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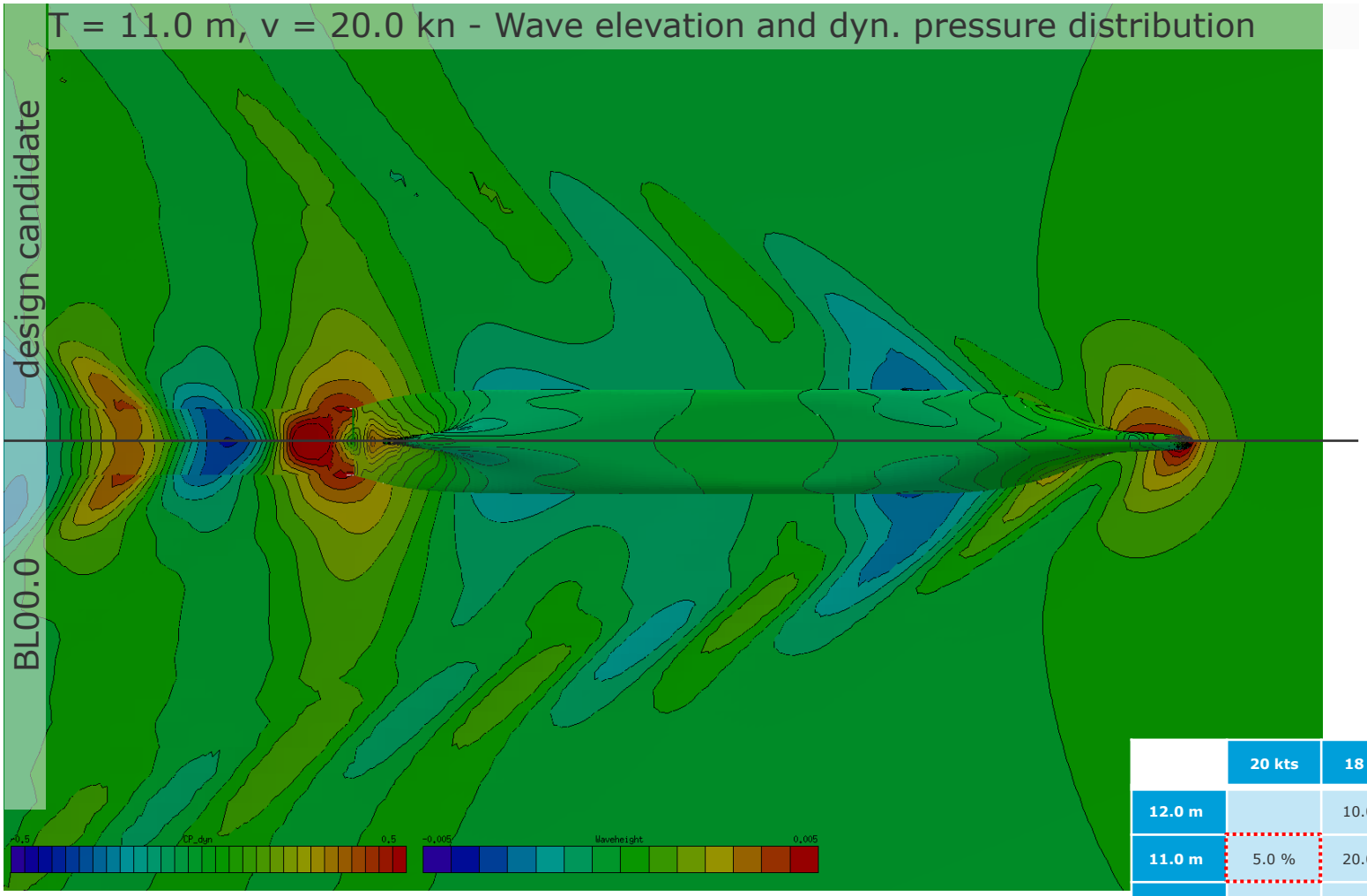

	20 kts	18 kts	16 kts	14 kts
12.0 m		10.0 %		10.0 %
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10.0 m				15.0 %

ECO-Retrofit Potential CV4300TEU

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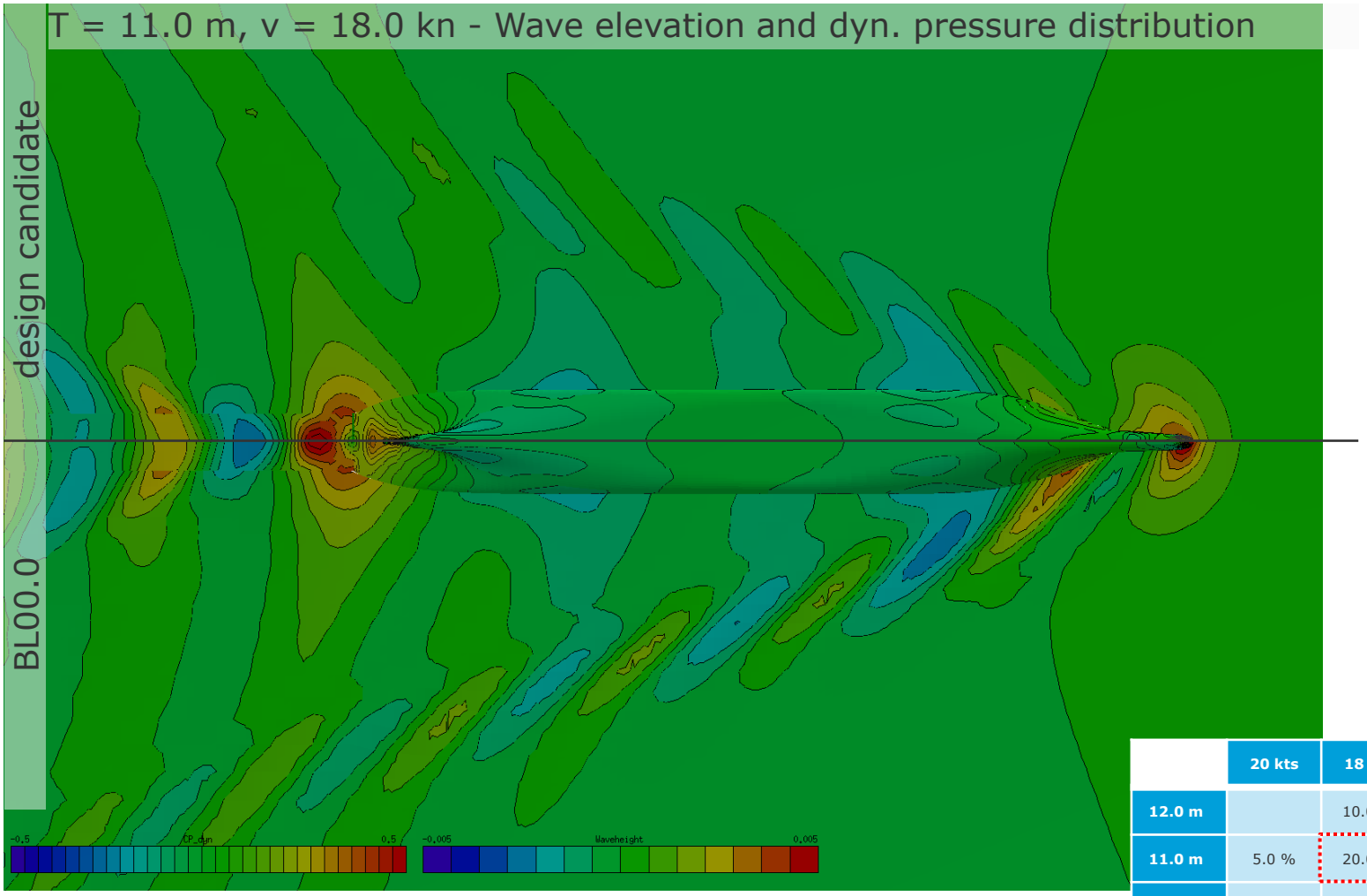

	20 kts	18 kts	16 kts	14 kts
12.0 m		10.0 %		10.0 %
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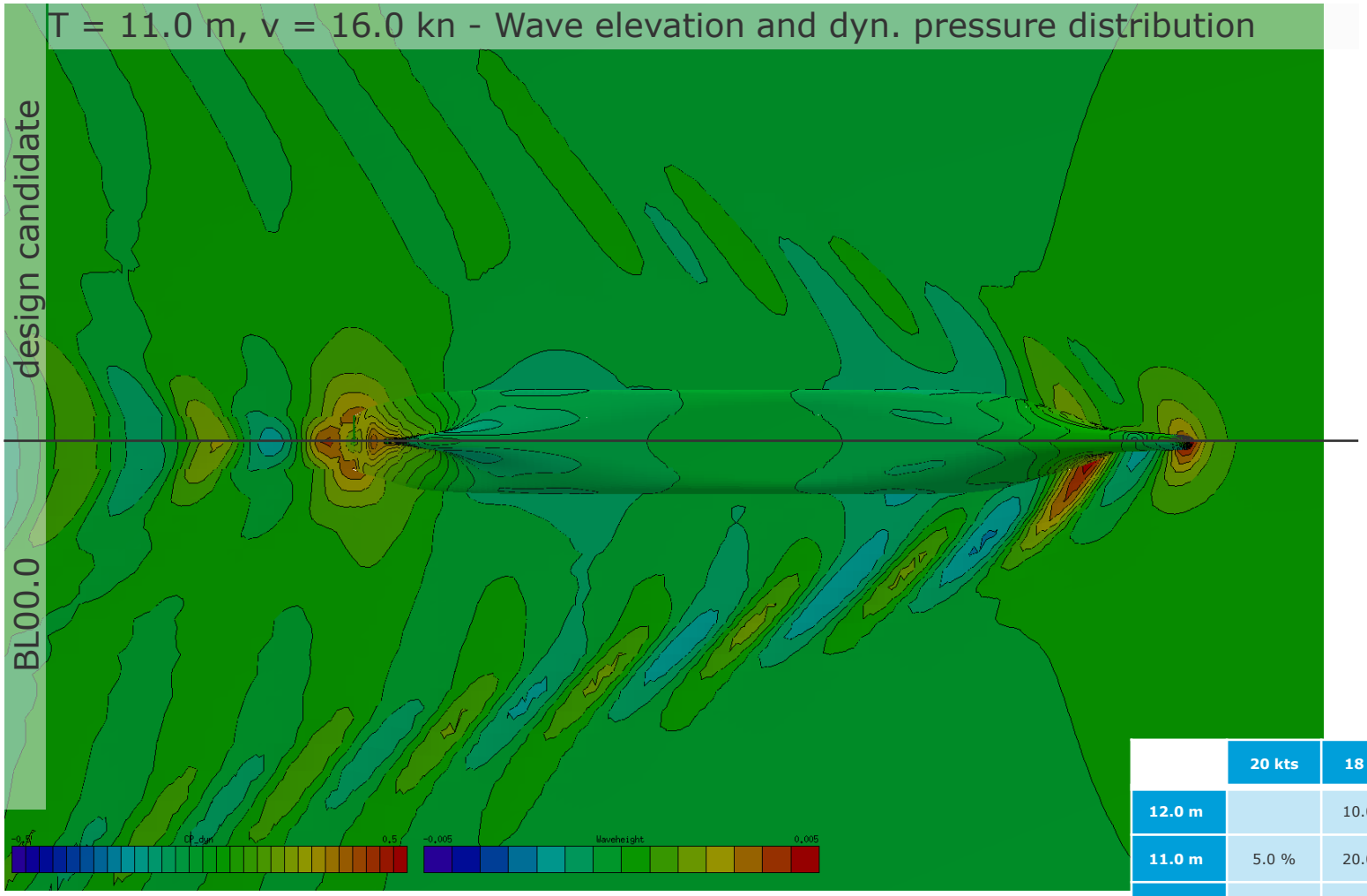
10.0 m				15.0 %

ECO-Retrofit Potential CV4300TEU

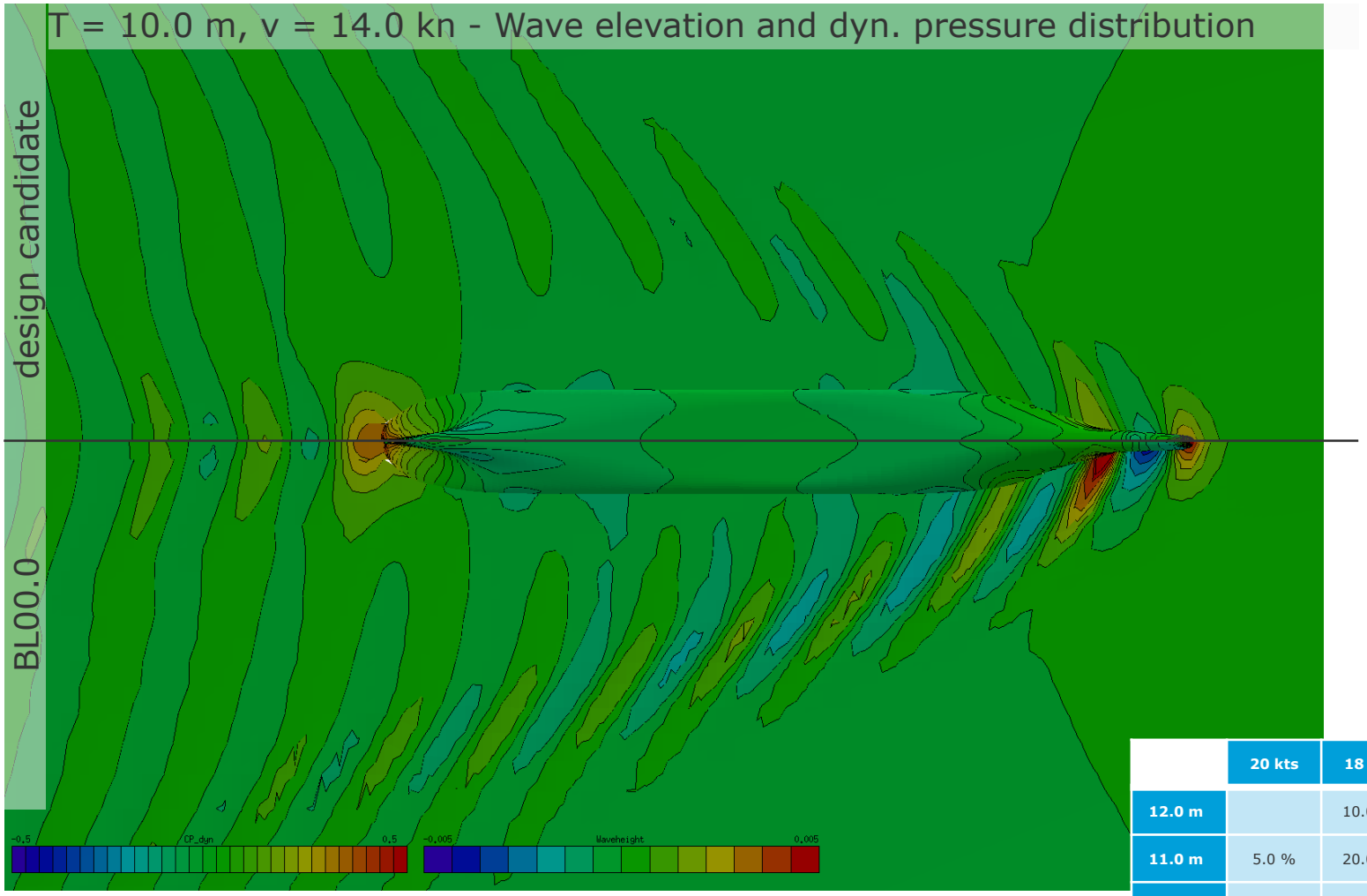
Flow field comparison of BL00.0 and exemplary design candidate



	20 kts	18 kts	16 kts	14 kts
12.0 m		10.0 %		10.0 %
11.0 m	5.0 %	20.0 %	40.0 %	
10.0 m				15.0 %

ECO-Retrofit Potential CV4300TEU

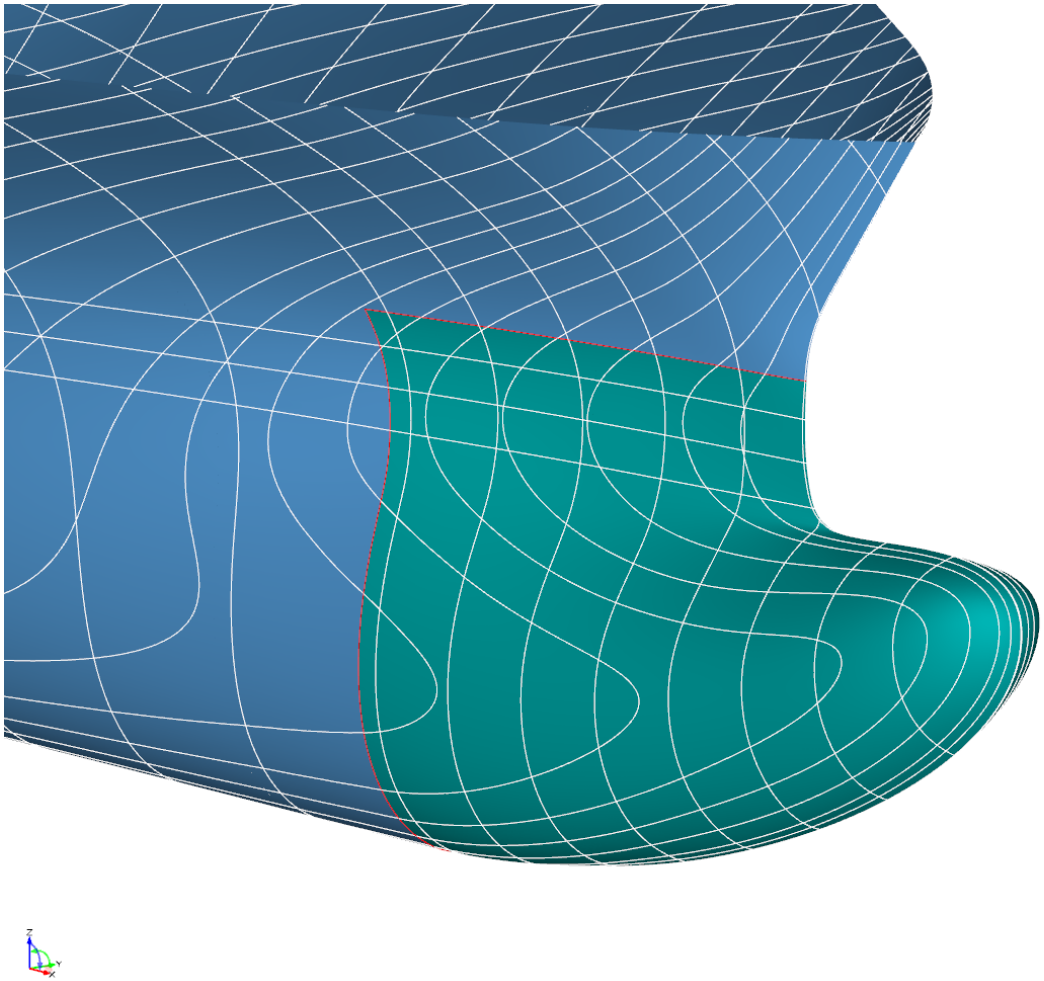
Flow field comparison of BL00.0 and exemplary design candidate



	20 kts	18 kts	16 kts	14 kts
12.0 m		10.0 %		10.0 %
11.0 m	5.0 %	20.0 %	40.0 %	
10.0 m				15.0 %

ECO-Retrofit Potential CV4300TEU

Impressions of an exemplary design candidate



ECO-Retrofit Potential CV4300TEU

Economical benefit estimation

Ship Type		CV 4300 TEU
Lpp	m	247
B	m	32.2
T	m	11
Installed power	kW	36560
Design speed	kn	24.3
NCR / MCR	%	90
sea margin	%	15

For information only – rough estimation of possible economic benefit based on weighted cases from optimization

Operational profile							
Ship speed	kn	18.0	14.0	20.0	18.0	16.0	14.0
Mean draft	m	12.0	12.0	11.0	11.0	11.0	10.0
Delivered power at trial condition	kW	12,046	5,668	15,593	11,368	7,984	5,019
Time at sea	260 days/year	10.0%	10.0%	5.0%	20.0%	40.0%	15.0%
	h/day	2.4	2.4	1.2	4.8	9.6	3.6

Engine	Fuel type	IFO380	IFO380	IFO380	IFO380	IFO380	IFO380
Specific fuel consumption	kg/kWh	0.176	0.176	0.176	0.176	0.176	0.176
Cylinder liner feed rate	g/kWh	0.800	0.800	0.800	0.800	0.800	0.800
Sludge	%	1.5%	1.5%	1.5%	1.5%	1.5%	1.5%

Savings (bow)	%	1.0%	6.5%	1.0%	3.5%	7.5%	15.0%	Total
Fuel oil savings	t/year	13.4	41.1	8.7	88.7	267.0	125.9	
Lubrication oil	t/year	0.06	0.18	0.04	0.40	1.20	0.56	
Fuel oil price	USD/t	550	550	550	550	550	550	
Lube oil price	USD/t	2500	2500	2500	2500	2500	2500	
Annual operational savings per vessel	USD/year	**7,536**	**23,047**	**4,877**	**49,779**	**149,833**	**70,648**	**305,720 USD/y**

Emmisions saved per year (approx.)								
Carbon dioxide (CO_2)	t/year	**41.8**	**127.7**	**27.0**	**275.9**	**830.3**	**391.5**	**1694.2 t/y**
SO_2	t/year	**1.07**	**3.29**	**0.70**	**7.10**	**21.36**	**10.07**	**43.6 t/y**
NOx	t/year	**0.00**	**0.02**	**0.00**	**0.03**	**0.10**	**0.05**	**0.2 t/y**

Investment cost and return on investment calculation

Wetted surface area of new bulbous bow	m²	463
Steel mass per wetted area	kg/m²	240
Costs per steel mass	USD/kg	6.5
Fleet size		3

Cost overview (approx)

- Optimization (bow)	flat	77,625 USD
- Model testing	flat	100,000 USD
- Steelwork for new bulbous bow	per vessel	722,280 USD
- New propeller	per vessel	0 USD
	Total (fleet)	2,344,465.00 USD
	Total (per vessel)	**781,488 USD**
	Return on investment	**30.7 months**

Financial assumptions	
Exchange rate €/$	0.80
Interest rate	6%
Years of operation	10

Accumulated savings after 10 years, 6.0% interest rate, fixed fuel price	USD	2,630,102 USD
Net present value when considering 10 years, 6.0% interest rate, fixed fuel price	USD	1,468,635 USD

ECO-Retrofit Potential CV4300TEU

ECO Retrofit Potential - Propulsion

an estimation of savings by refitting a new propeller and a new bulbous bow

Retrofit of a new propeller

- To evaluate the potential gains in power demand for a new propeller design three different scenarios have been investigated:

  At the targeted slow steaming profile and considering the decreased power demand due to an retrofitted bulbous bow section

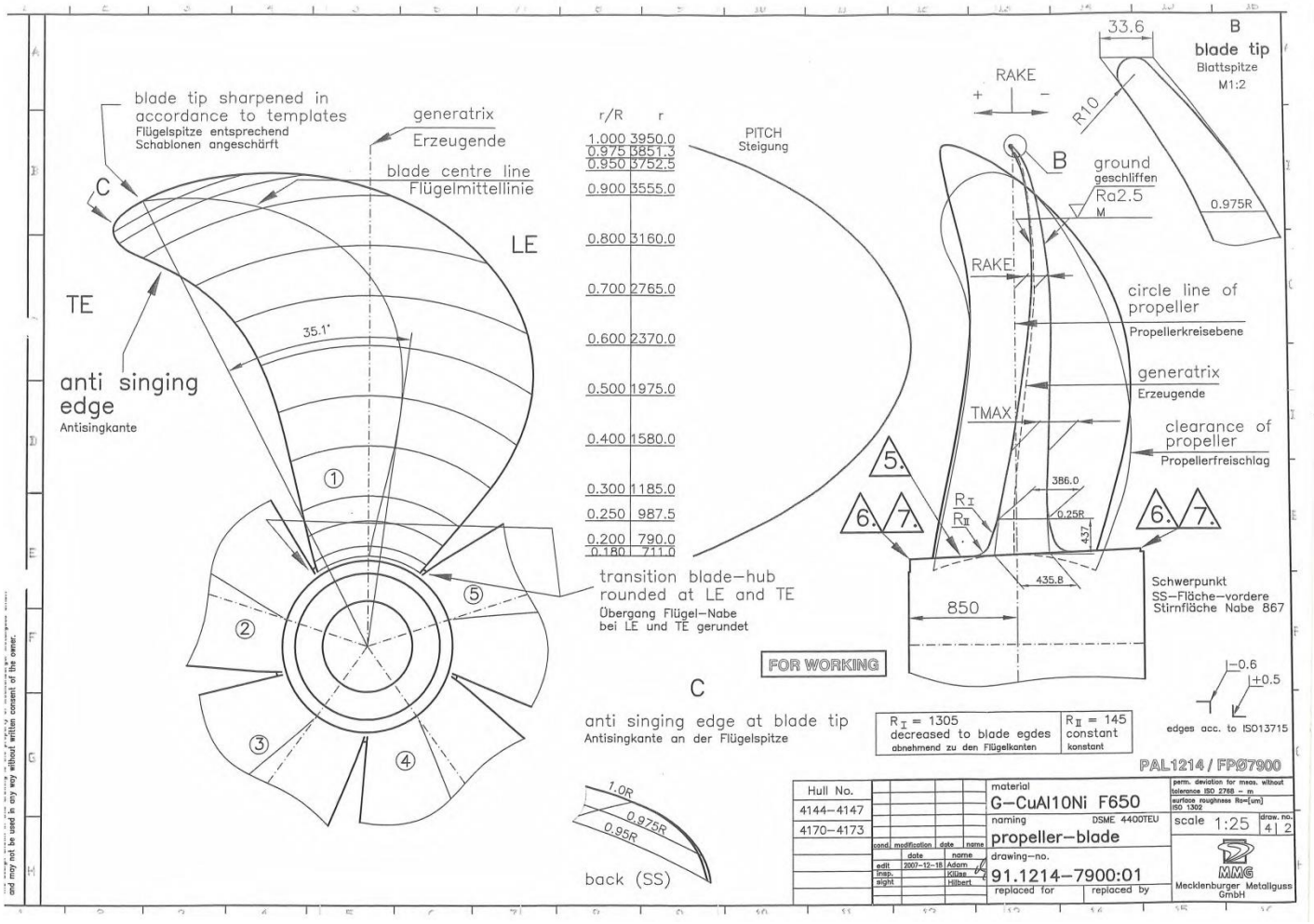
  1) The existing propeller design and engine setup
  2) A new propeller design; existing engine setup (no de-rating or MCR limiting measures)
  3) A new propeller design; adopted engine setup (de-rating or MCR limiting to SMCR=60% MCR)

- The potential improvements due to an updated propeller design are determined as the difference in power demand of scenarios 2) and 3) compared to scenario 1).

- For all scenarios the power demand has been estimated based on the total resistance as derived by means of CFD and empirical assumptions for hull efficiency (ETAH) and relative rotative efficiency (ETAR).

- Both new propeller designs have been chosen such, that pressure pulses do not exceed the values of the existing propeller.

Drawing of existing propeller



ECO-Retrofit Potential CV4300TEU

Potential new propeller design / existing engine setup

Savings in Total Resistance $PD_{service}$	A (20 kn)	B (18kn)	C (16 kn)	D (14 kn)
A (12.0 m)		1.8 %		1.8 %
B (11.0 m)	1.8 %	1.8%	1.8 %	
C (10.0 m)				1.8 %

- A preliminary propeller design has been carried out under the assumption that no de-rating / limiting measures are intended (meaning the engine is capable of the full MCR and ship speed exceeding the intended new maximum speed of 20kn can be reached).
- The propeller diameter was slightly increased to 8m.
- Thus the potential gain of a propeller retrofit results to a weighted average of approx. **1.8%** compared to the existing propeller.

Potential new propeller design / adopted engine setup

Savings in Total Resistance $PD_{service}$	A (20 kn)	B (18kn)	C (16 kn)	D (14 kn)
A (12.0 m)		3.8 %		3.8 %
B (11.0 m)	3.8 %	3.8%	3.8 %	
C (10.0 m)				3.7 %

- A preliminary propeller design has been carried out under the assumption that de-rating / limiting measures are applied. For this investigation SMCR is limited to 22,000 kW (approx. 60% MCR). The propeller rotation speed is chosen on the existing propeller power curve.

  It has been verified with the license owner of the engine (MAN) that simple power limiting measures (e.g. turbo charger cutout with by-pass) will be applicable to achieve this reduction. Detailed measures however have to be checked with the engine manufacturer.
- The propeller diameter was slightly increased to 8m.
- Considering an adopted engine setup the potential gain of a propeller retrofit results to a weighted average of approx. **3.8%** compared to the existing propeller.

Comparison of propeller main particulars

	Existing propeller	Future propeller (1)	Future propeller (2)
Diameter	7.9m	8.00m	8.00m
Blade no.	5	5	5
Mean pitch ratio	0.9850	0.9606	0.9668
Exp. blade area ratio	0.7269	0.7201	0.5920
Mass	53.2t	55.9t	46.8t

Note: The future propeller variants have been designed with the same boss dimensions as the existing propeller.

Economical benefit estimation

Ship Type		CV 4300 TEU
Lpp	m	247
B	m	32.2
T	m	11
Installed power	kW	36560
Design speed	kn	24.3
NCR / MCR	%	90
sea margin	%	15

For information only – rough estimation of possible economic benefit based on weighted cases from optimization

Operational profile							
Ship speed	kn	18.0	14.0	20.0	18.0	16.0	14.0
Mean draft	m	12.0	12.0	11.0	11.0	11.0	10.0
Delivered power at trial condition	kW	12,046	5,668	15,593	11,368	7,984	5,019
Time at sea	260 days/year	10.0%	10.0%	5.0%	20.0%	40.0%	15.0%
	h/day	2.4	2.4	1.2	4.8	9.6	3.6

Engine	Fuel type	IFO380	IFO380	IFO380	IFO380	IFO380	IFO380
Specific fuel consumption	kg/kWh	0.176	0.176	0.176	0.176	0.176	0.176
Cylinder liner feed rate	g/kWh	0.800	0.800	0.800	0.800	0.800	0.800
Sludge	%	1.5%	1.5%	1.5%	1.5%	1.5%	1.5%

Savings (bow + propeller)	%	4.8%	10.3%	4.8%	7.3%	11.3%	18.8%	
Fuel oil savings	t/year	64.5	65.1	41.7	185.0	402.3	157.8	
Lubrication oil	t/year	0.29	0.29	0.19	0.83	1.80	0.71	
Fuel oil price	USD/t	550	550	550	550	550	550	
Lube oil price	USD/t	2500	2500	2500	2500	2500	2500	Total
Annual operational savings per vessel	USD/year	**36,172**	**36,521**	**23,411**	**103,824**	**225,748**	**88,545**	**514,222 USD/y**

Emmisions saved per year (approx.)								
Carbon dioxide (CO_2)	t/year	**200.5**	**202.4**	**129.7**	**575.4**	**1,251.0**	**490.7**	**2849.7 t/y**
SO_2	t/year	**5.16**	**5.21**	**3.34**	**14.80**	**32.18**	**12.62**	**73.3 t/y**
NOx	t/year	**0.02**	**0.02**	**0.02**	**0.07**	**0.15**	**0.06**	**0.3 t/y**

Investment cost and return on investment calculation

Wetted surface area of new bulbous bow	m²	463
Steel mass per wetted area	kg/m²	240
Costs per steel mass	USD/kg	6.5
Fleet size		3

Cost overview (approx)

- Optimization (bow + propeller)	flat	77,625 USD
- Model testing	flat	100,000 USD
- Steelwork for new bulbous bow	per vessel	722,280 USD
- New propeller	per vessel	300,000 USD
	Total (fleet)	3,244,465.00 USD
	Total (per vessel)	**1,081,488 USD**
	Return on investment	**25.2 months**

Financial assumptions	
Exchange rate €/$	0.80
Interest rate	6%
Years of operation	10

Accumulated savings after 10 years, 6.0% interest rate, fixed fuel price	USD	4,841,073 USD
Net present value when considering 10 years, 6.0% interest rate, fixed fuel price	USD	2,703,230 USD

ECO-Retrofit Potential CV4300TEU

ECO Retrofit Potential – Executive summary

Conclusion

- For a **bulbous bow retrofit potential savings** on total resistance (Rt) have been assessed to **approx. 6.8%** weighted average over the intended operational profile of the vessel. Higher savings of up to 15% are expected for the lower speeds and lower drafts while on higher speeds and higher drafts savings are moderate.
- For a **propeller retrofit potential savings** have been assessed to approx. 1.8% (without any engine de-rating / limiting measures) and **3.8%** (when limiting main engine MCR) throughout the matrix.

Please note:

The savings as assessed during the potential phase of the ECO-Retrofit Bow and Propeller might be subject to small changes during the optimization as applied in the second phase.

Thank you for the good cooperation!

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